



Full Length Article

Challenges of Forestry Management in Nasarawa State, Nigeria

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ABSTRACT

The paper looked at challenges of forestry management in Nasarawa state. It took cognizance of the management status of the forest estate, agencies involved in forest management, manpower situation, budgetary appropriation and revenue generation for five years (2002-2006) among others. Both primary and secondary data were collected, the former were collected using structured questionnaire and interpersonal interview. The study showed that the forest is an important repository for meeting man's economic and welfare need as depicted by respondents clamour for medicinal uses (43.3%); farming and other economic activities (39.9%); food and other forest products (12.6%); religion/idol worship (1.6%); hunting (1.1%); political/social (1.1%); and water supply (0.8%). The study also revealed that Nasarawa state has 41 forest reserves covering 0.145 million ha with only 20 forest guards giving a ratio of 1:7261 coverage, which is ineffective. Forest plantation span is 41 years (1966-2007) with total area of 5,537.14 ha averaging 135.05 ha/annum, which is very low to meet the wood requirements of the state; the only proposed game reserve in the state is yet to function; the total budgetary appropriation between 2002-2006 was ₦ 186.150 million (\$1.538 million) and only ₦ 2.00 million (\$0.17 million) was released during this period; revenue generated stood at ₦ 6.58 million (\$0.54 million), none of which was ploughed back into forestry development; six agencies were involved in forest management, this should be streamlined for effectiveness; the manpower situation of 2 professional staff, 11 technical staff and 24 vocational staff is definitely not adequate. Generally, for forestry to be developed in Nasarawa state, all these shortcomings as revealed by this study should be addressed promptly.

Key Words: Challenges; Forestry management; Revenue generation; Nasarawa state; Nigeria

INTRODUCTION

The contribution of forestry to national development cannot be overemphasized. Forest supplies the resources for furniture making, veneer, plywood, sawn timber, particle board, pulp and paper, paper board, wood for energy (charcoal & fuel wood), gum, resin, ropes, fruits, bush meat, etc. However, demand for timber far outstrips the supply (Alao, 2003). Consequently, wood shortage is imminent in the country. Nwokeabia (1983) showed that demand for sawn wood alone on the basis of current consumption is estimated to be 1.2 million m³ per year. This is expected to grow by 6-9% annually. The question now is, with the wanton destruction of the forest: can this demand be met and sustained? This is aptly put in focus by Nnabuike (2003) who warned that people cut trees indiscriminately; they do not know that one day, we are going to run out of these trees and it is already happening in some countries.

According to Onochie (1984) government's response to the need of forestry development has not been forthcoming as it should be, that it has not been accorded a high enough priority considering the danger, which the nation faces in the breakdown in its wood supply. Alao (2005a) also opined that all governments (Federal, State & Local) in Nigeria are not responsive to the issue of

sustainable forestry development in the country.

Forestry is a very complex enterprise indeed. Its most distinctive feature is that it is invariably a multi-product enterprise, some of the forest products such as timber are physical and have market determined values (Alao, 2005b). Because of this and changing needs of the society, forestry development is very imperative. This paper therefore highlights the progress of forestry development in Nasarawa state and proffer appropriate solutions accordingly.

MATERIALS AND METHODS

Study area. Nasarawa state was created in 1996. It has a land area of 27,116.80 square kilometers. The state lies between latitude 8°54'N and longitude 8° 39'E. The mean annual rainfall is 1550 mm and mean monthly temperature is between 20°C and 34°C.

Data collection. Both primary and secondary data were collected and used for this study. The primary data were collected from the Forestry Department, Ministry of Agriculture and Natural Resources, Lafia, Nasarawa state using structured questionnaire and interpersonal interview. The primary data collected include number of forest reserves and their status; land area; manpower position; status of artificial plantation development; budgetary

appropriation (2002-2006); revenue generation (2002-2006); and the various agencies involved in forest management in Nasarawa state. While the secondary data were taken on the major forest uses.

RESULTS AND DISCUSSION

Major uses of forest. Forest uses are wide and far reaching. For instance in a survey by Nweze (1997) covering 365 respondents in Ebonyi and Imo States, 43.3% were of the perception that forests provide the habitat for biodiversity, which are considered to have medicinal value. While following closely in importance was farming with 39.9%. Allied to these were the value of forests as a source of food and other forest products such as fruits, nuts, latex, honey, rattan, meat and oils. Also notable, albeit with lower responses were the use of forests as sources of water supply, religious, political and social activities (Table I).

Obviously, these are economic functions, which directly or indirectly contribute to the welfare of communities in which the forests are situated. We can see that the first priority of the respondents was the use of the forest for medicinal purposes which are non-timber forest products. This has gone to show that lack of sustainable development of our forests will surely cause the genetic erosion of these valuable resources.

Forestry management in nasarawa state. Forestry management practice in Nasarawa state is in two categories. We have the artificial plantation otherwise called plantation forestry and natural regeneration. The total land areas of Nasarawa State under forest reserve cover is 150,765.26 ha, which is just 5.6% of the total land area of the state. Even this 5.6% forest cover is threatened by environmentally unfriendly attitudes like illegal felling, farming, bush burning, encroachment, dereservation, etc and management incapacitation hinged on government apathy. This percentage is too less from the national minimum forest cover of 25%. There is the need to put up a concerted effort to increase the forest cover of the state.

Agencies involved in forest management in nasarawa state. There are six (6) identified organizations managing forest in the state in one form or the other. The agencies are Nasarawa State Agricultural Development Project (NADP); Nasarawa State Urban Development Agency; Local governments through their Departments of Agriculture; state government through the Department of Forestry, Ministry of Agriculture and Natural Resources; Ministry of Water Resources and Rural Development; and Nasarawa State Environmental Protection agency.

These multiplicities of agencies has engendered conflict in forest management in Nasarawa state and this does not augur well for the proper and effective management and development of the forest resources of the state more so that all of them are government agencies. The only body that should be allowed to manage and superintend over forest resources in the state is the

Table I. Major Uses of Forests

Uses	No of Responses	Percentage (%)
Medicinal	158	43.3
Farming and other economic activities	144	39.9
Food and other forest products	46	12.6
Religion/Idol worship	6	1.6
Hunting	4	1.1
Political/social	4	1.1
Water supply	3	0.8
Total	365	100.0

Source: Nweze, 1997

Table II. Natural Forest Reserves in Nasarawa state

Name of Forest Reserve	Size (Ha)	Legal Status	Degree of Protection
Karama	291.84	Gazetted	Not protected
Utuga	677.20	"	"
Gwanje	20.48	"	"
Righa	40.96	"	"
Kurmin Tagyaye North	25.60	"	"
Kurmin Tagyaye South	28.16	"	"
Mada River North	3,013.12	"	"
Mada River South	1,845.70	"	"
Kurmin Agyaga	50.00	"	"
Kurmin Nunku	179.12	"	"
Kurmin Nunkuchu	56.32	"	"
Kanje	200.00	"	"
Rafin Pa (Azara)	156.16	"	"
Doma	86,374.40	"	"
Ambakar	15,462.40	"	"
Marhai	6,883.84	"	"
Kurmin Agyaragu	43.52	"	"
Kurmin Agudu	23.04	"	"
Kurmin Akanga	2,808.70	"	"
Kyakale	22.00	"	"
Adevi	213.00	"	"
Baba	122.00	"	"
Akaleku	102.00	"	"
Doka	1,800.00	"	"
Keffi Town (Sabon Gari)	181.76	"	"
Dustsen Karaga	2,508.80	"	"
Keana	-	Proposed	"
Kaffa Kurmi (Kugbaru)	194.56	Gazetted	"
Agudu	161.28	"	"
Kurmi Mai Akuya	568.08	"	"
Arikya	20.00	"	"
Kurudu	100.00	Proposed	"
Bakyano	-	"	"
Obi	2,040.32	Gazetted	"
Atabla	150.00	"	"
Zano	1,088.00	"	"
Bakono (Game Reserve)	12,160.00	Proposed	"
Gitata/Jaja	3,417.60	Gazetted	"
Sheriga (Buku)	27.28	"	"
Tokan River	1,868.80	"	"
Wamba	302.08	"	"
Total	145,228.12		

Source: Ministry of Agriculture and Natural Resources, Lafia, Nasarawa state

Department of Forestry, Ministry of Agriculture and Natural Resources, but paying royalties to local governments, where forest resources are exploited appropriately.

Natural forest reserve. This covers an area of 145,228.12 ha. The common species in these reserves are *Daniellia oliveri*, *Khaya senegalensis*, *Sterculia setigera*, *Vitex*

Table III. Forest Plantation Estate of Nasarawa state

Location (Local Government/Development Area)	Year of Establishment	Total Area Planted (Ha)	Species Planted
Akwanga	1975	40.0	<i>Gmelina arborea</i> <i>Tectona grandis</i>
Doma	1977-1981	1,859.0 715.0	<i>Gmelina arborea</i> <i>Tectona grandis</i>
Farin Ruwa	2006	155.7	<i>Gmelina arborea</i> <i>Tectona grandis</i>
Jenkwe	1966-1982	1,898.0 548.7	<i>Gmelina arborea</i> <i>Tectona grandis</i>
	1980-1981	23.04	<i>Gmelina arborea</i> <i>Tectona grandis</i> <i>Eucalyptus spp</i>
	1980-1982	40.0	<i>Tectona grandis</i> ; <i>Eucalyptus spp</i>
	2007	15.0	<i>Gmelina arborea</i> <i>Tectona grandis</i>
Keffi	1980	56.7	<i>Gmelina arborea</i> <i>Tectona grandis</i>
	2007	35.0	<i>Gmelina arborea</i> <i>Tectona grandis</i>
Lafia	1981-1982	31.0	<i>Tectona grandis</i>
	-	20.0	<i>Eucalyptus spp</i>
Nasarawa	-	100.0	<i>Gmelina arborea</i> <i>Tectona grandis</i>
Total		5,557.14	

Source: Ministry of Agriculture and Natural Resources, Nasarawa state

doniana, etc. These reserves are scattered virtually all over the state.

There are forty one (41) forest reserves in the state, out of these 37 are gazetted, while 4 are ungazetted. Their management status shown by protected and not protected is shown in Table II. Though, these reserves ought to be protected and managed sustainably, the reverse is however the case. There is free entry and exit into these reserves due to inadequate number of forest guards to patrol and secure them against illegal entry and activities. The state has only twenty (20) Forest guards to police a forest reserve area of 145,228 12 ha. This is a ratio of 1:7261 i.e., 1 forest guard to police 7,261 hectares. This is grossly inadequate and something should be done in order to protect the forest resources of the state. The irony of it all is that, if the forest reserves are not protected what then become of the free areas – areas outside the reserves that are not legally constituted. Certainly unrestrained exploitation and degradation will have a free reign.

Game reserve. It is very disheartening to point out that Nasarawa state has no single game reserve. The proposed Bakono game reserve covers an area of 12,160 ha (Table II). It is yet to be legally constituted: as a result the fauna and the flora within this enclave are exposed to poaching, massive exploitation and degradation.

Artificial plantation estate. The total plantation area in Nasarawa state is 5,537.14 ha (Table III). Forest plantation establishment started in the state in 1966 up to 2007. This is a span of 41 years. One can therefore infer that an average area of 135.05 ha/annum was planted between 1966 and 2007. This is very dismal indeed. This definitely portends grave danger to the wood security of the state. The species generally planted were *Gmelina arborea*, *Tectona grandis* and *Eucalyptus* spp.

It is sad to state that all the *Tectona grandis* were harvested for export between 1995 and 1998. Money realized from stumpage sales were not ploughed back either to expand the forest plantation estate or maintain the coppices. The paradox now is that the coppices that lived through the “survival of the fittest” are today being harvested and sold either as transmission poles or thinning by the same government that have refused to plough money back into their maintenance.

Manpower. Manpower is the human resources needed to leverage any organization. The state has only 2 professional staff, 11 technical staff and 24 vocational staff made up of 20 forest guards and 4 range guards (Table IV) This is grossly inadequate to manage the forest estate of the state covering a combined total area of 150,765.26 ha (both natural forest reserve & forest plantation) and carry out any other meaningful development plans.

Budgetary appropriation. Budgetary appropriation to forestry was merely a yearly ritual, because at the end of the day money was never released by the government for its intended purpose. In a period of five (5) years (2002-2006) it was only in 2006 that a paltry sum of ₦2.00 million (8.51%) was released out of the budgeted sum of ₦23.50 million (Table V). The total money appropriated to forestry between 2002 and 2006 stood at ₦18.16 million and only ₦2.00 million that is 1.07% was released. Funding was a bane in realizing both forestry and institutional development by the forestry department. Government’s apathy to the funding of forestry and wildlife development as against prestigious projects, which yield quick returns could perhaps be as a result of their long-term nature, which does not yield tangible or immediate benefits. There is however, the need for adequate funding in order to achieve meaningful forestry development in the state.

Table IV. Manpower Position as at 2007

Professional cadre	Technical cadre	Vocational cadre
2	11	24

Source: Forestry Department, Ministry of Agriculture and Natural Resources, Lafia, Nasarawa State

Table V. Budgetary Appropriation between 2002 and 2006

Year	Budgetary appropriation ₦ (Million)	Amount Released ₦ (Million)	Percentage (%)
2002	62.850 (\$0.524 Mil)	Nil	0
2003	33.400 (\$0.278 Mil)	Nil	0
2004	33.900 (\$0.282 Mil)	Nil	0
2005	32.500 (\$0.277 Mil)	Nil	0
2006	23.500 (\$0.196 Mil)	2.00 (\$0.017 Mil)	8.51
Total (2002-2006)	186.150 (\$1.551 Mil)	2.00 (\$0.17 Mil)	1.07

Source: Forestry Department, Ministry of Agriculture and Natural Resources, Lafia, Nasarawa State

Table VI. Revenue Generation by the Forestry Department covering 2002 to 2006

Year	Revenue Generated; ₦ (Million)
2002	1,680,721.00 (\$0.014 million)
2003	1,314,725.00 (0.011 million)
2004	1,231,130.00 (\$0.010 million)
2005	1,127,145.00 (\$0.009 million)
2006	1,230,250.00 (\$0.010 million)
Total (2002-2006)	6,583,971.00 (\$0.54)

Source: Forestry Department, Ministry of Agric & Natural Resources, Lafia, Nasarawa State

Revenue generation. Total revenue generated between 2002 and 2006 stood at ₦6.581 million (Table VI) as against only ₦2.00 million released during the period under review. This revenue so collected was not ploughed back into forestry development but paid into the government treasury from where the money went into something else other than for forestry use.

CONCLUSION AND RECOMMENDATIONS

Forestry development in Nasarawa state faces a very bleak future. There is an urgent need to rebuild and restore the depleting resources. The forestry service of the state can deliver if given adequate funding and the engagement of appropriate manpower.

Tree planting is an expensive operation, which has to be undertaken only when it is absolutely necessary. Forestry development in Nasarawa state is an absolute necessity. To this end the following recommendations are suggested:

- There should be a political will on the part of government to confront issues bordering on forestry development in Nasarawa state in an absolute term.
- Budgetary allocation to forestry should always be released as at when due because forestry operations are time bound.

- The forestry department should be allowed to share out of the revenue they generate so that this can be ploughed back into forestry development.
- In order to further mitigate funding problems forestry trust fund (FTF) could be established.
- The norm for sustainable conservation is that for every tree removed there must be adequate replacement. For this to be effective a system known as regeneration tax should be introduced. This should be built into the stumpage price paid by timber contractors. The revenue accruing from this tax should be handed over to the forestry department for forestry development purposes.
- The Nasarawa state Government should explore the possibility of upgrading the forestry department into a forestry commission. This will confer autonomy and access to international funding for forestry development.
- Forestry management in Nasarawa state should be streamlined.
- Adequate manpower in quality and quantity should be engaged especially those needed to police the forest reserves.
- Government should put in place the machinery to establish and nurture the Bakono game reserve into an ecotourism centre. This will generate revenue for the state.
- The government should explore the possibilities of community participatory management strategy. This is a kind of confidence building tool between the government and the communities where these forest estates are situated.

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